Eddie Tang

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Education

University of Illinois Urbana-Champaign Urbana, IL
*BS in Electrical Engineering | GPA: **4.0*** *Aug. 2025 – May 2029*
Minor in Computer Science

Projects & Leadership

CAM-MK3 | Illinois Space Society - Spaceshot Avionics | KiCAD, C++, ESP32-IDF Oct. 2025 – Aug. 2026

- Created compact live-video transmitter & dual-camera controller PCBA for rocket avionics; designed schematic, layout, and power architecture to meet size and RF link budget constraints.
- Authored and validated firmware drivers for NTSC→digital video, hardware JPEG encoding, & radio transceiving
- Implemented B2B I^2C control and 433 MHz wireless communication for telemetry commands and live-video feed
- Built and debugged multiple revisions and validated signal & power integrity with oscilloscope and bench power supplies. Flown built boards on rockets up to 13,455ft and 30,994ft.

Project EMBER (Technical Lead) | *Illini Solar Car* | KiCAD, C++, Simplicity Studio Jan. 2026 – Jul. 2026

- Spearheaded Project EMBER, a joint initiative with AbbVie to investigate ultra-low-power design and implementation for space-constrained PCBs, through the development of a TPMS system.
- Facilitated design, development, & bring-up of schematic and layout of a 17mm-diameter TPMS sensor node PCB
- Developed custom FSM, power system, and BLE payload to consume 420nA when idle and 15µA average when active, allowing continous use of up to 283 days.
- Coordinated deliverables, weekly meetings, KPIs, timeline, and project reports with AbbVie by delegating work across an 8-member team spanning firmware and receiver node layout.

GATOR | Illinois Space Society - Spaceshot Avionics | KiCAD, C++ Oct. 2025 – Present

- Implemented & designed GATOR, a power-distribution PCB to drive 24V servos to dynamically control the orientation of a yagi antenna according to relative rocket position.
- Programmed GATOR orientation feedback loop using GPS data from LoRa telemetry packets and servo angle

TARC Avionics V1–V3 (Lead & Founder) | EasyEDA, C++, Python Jun. 2022 – Jun. 2025

- Designed, assembled, and programmed custom 6-DOF AHRS PCB using BMI088 IMU and MS5607 barometer for active flight control to variable target apogees with 80% success rate
- Developed quaternion-based sensor fusion algorithm for accurate altitude & velocity estimation from acceleration
- Built Python software-in-the-loop simulations to validate control logic using historical flight telemetry
- Ranked top 50 at TARC Nationals (from pool of 1,000 teams); qualified twice for Nationals

Experience

Avionics Lead | Prev. Launch Ops Member | *Illinois Space Society - Spaceshot Avionics* Jun. 2025 - Present

- Leading 4 subteams (Ehardware, Software, AV-structures, GNC), managing over 12 PCBs & respective code bases
- Manned, assembled, and configured flight ground stations for 4 high-power rocket launches (>20k ft), including LoRa/RF link verification, telemetry receiver initialization, and antenna pointing before each launch window.
- Conducted pad procedures & AV-bay integration, including continuity checks on pyros and arming sequences
- Monitored real-time telemetry data during flight and recovery phases to support post-flight vehicle recovery.

Research Intern, DOE SULI Program | *Pacific Northwest National Laboratory* May 2026 - July 2026

- Engineered low-power RF transmitter for RF bat tags by implementing novel design and firmware for FSK modulation of Gold code via programmable oscillators; Successfully field tested up to 324 m.
- Designed PCB in Altium to test two designs (with and without RF switch) to evaluate signal-integrity tradeoffs.
- Achieved 12 dBm output power into a 50Ω load. Characterized RF designs using a spectrum analyzer and VNA.
- Authored research manuscript and delivered poster presentation on transmitter design and characterization.

Research Assistant (Paid), Mechanical Engineering | *Rowan University* Summer 2025

- Investigated effects of point defects in thermoelectric materials on lattice thermal conductivity using MD.
- First author on accepted manuscript investigating defect effects on thermal conductivity in Si and Ge.

Laboratory Learning Program, Mechanical & Aerospace Department | *Princeton University* Summer 2024

- Performed DFT calculations on WO_3 surfaces for plasma-assisted NH_3 synthesis; analyzed results with NumPy.
- Co-authored a peer-reviewed publication in ACS Energy Letters (Impact Factor: 19.3).

Skills

Design Tools: KiCAD, Altium, EAGLE, Vivado, OnShape, NX Siemens, Git, LAMMPS, VASP, Slurm
Languages: C++, Python, C, Bash, Assembly, Dart, Go, Java, MATLAB
Protocols: I2C, SPI, UART, CAN, USB, Bluetooth, WiFi, LoRa, FSK
Lab Equipment: Oscilloscope, Function Generator, Spectrum Analyzer, Software Defined Radio, RF Amplifier
Other: HAM Radio Technician (KE2CNQ); Fluent in English and Chinese; NMSQT Finalist; 1560 SAT